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THE VISCOSITY OF SUSPENSIONS OF SPHERES

JAMES V. ROBINSON
The Mead Corporation, Chillicothe, Ohio

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INTRODUCTION

Several equations to express the viscosity of suspensions have been proposed, most of which involve arbitrary constants "characteristic of the materials." The Arrhenius equation is

$$\eta = \eta_0 e^{kc}$$

in which k is empirical and c is the weight of solute per cubic centimeter of solvent (1). Weltmann and Green (12), modifying the Arrhenius equation, proposed that

$$U = (\eta_0 + A)e^{Bp}$$

in which U is the plastic viscosity, p is the volume per cent of solid, and A and B are constants of the materials. Einstein (4) derived theoretically the relationship for the viscosity of suspensions of spheres at low concentration,

$$(\eta - \eta_0)/\eta_0 = \eta_{\text{sp}} = kV$$

and later found the constant to be 2.5. Eirich, Bunzl, and Margaretha (5) experimentally confirmed the Einstein equation for concentrations of spheres up to 5 to 10 per cent by volume, depending on the viscosimeter used.

Hatschek (8) proposed a theoretical law in the same form as the Einstein equation, with a constant of 4.5 instead of 2.5, applicable to a concentration of 40 per cent by volume of dispersed phase, and another equation applicable at high concentrations: $(\eta - 1)/\eta = V^{1/3}$. Guth and Simha (7) derived an equation for solutions and suspensions, in which the relative viscosity, η/η_0, is expressed as a power series of the volume concentration of the suspended material. Vand (11) more recently made a theoretical derivation which he reduced to a similar power series expressing the relative viscosity; this equation fitted his experimental results fairly well at low concentrations but deviated above approximately 25 per cent by volume concentration. None of these equations have been applicable over the entire range of suspension concentration. Norton, Johnson, and Lawrence (10) have set up the equation

$$\eta = \eta_0(1 - c) + k_1c + k_2c^n$$

and have empirically determined the parameters, by which the flow behavior of clay suspensions may be characterized, at any concentration.

As further indicated by Bredée and de Booys (2), the relationships which are valid over a wide range of concentration involve parameters without direct physical significance. An attempt is made herein to demonstrate a relationship with two parameters, one of which has simple physical significance, valid over the entire range of suspension concentration.

DERIVATION

The Einstein equation states that at low suspension concentrations the specific viscosity, $(\eta - \eta_0)/\eta_0$, is directly proportional to the volume concentration of the suspended solid, $\eta_{sp} = kV$. This equation was derived for very dilute suspensions, in which the volume occupied by the spheres was negligible. To extend the equation to higher concentrations, the concept is introduced that the specific viscosity is not only proportional to the volume concentration, but that it is also inversely proportional to the volume of free liquid in the suspension.

The volume of "free liquid" in the suspension is that liquid outside of the suspended particles; in case the particles are highly aggregated, there will be liquid entrapped within the aggregates which will contribute nothing to the fluidity. The free liquid therefore is not the difference between the total volume and the actual volume of solid, but that between the total volume and an effective volume of the solid. What this effective volume is may be deduced from a consideration of the upper concentration limit of the suspension. When the concentration of solid becomes sufficiently high, a point is reached at which each particle is in firm contact with its neighbors, that is, there is just sufficient liquid to fill the voids between the packed particles. Under such conditions fluid flow is impossible; unless sufficient stress is applied to change the condition of the solid particles, the "suspension" has become a porous solid and its viscosity is infinite. Under these conditions the "suspended" particles constitute a packed sediment, whose specific volume will depend upon the particle-size distribution and the degree of aggregation. The effective volume of the particles is then the packed-sediment volume; the data presented below show that the packed-sediment volume appears to be the effective volume of the particles at any concentration.

The sediment volume is expressed as "relative sediment volume" (S'), that is, the volume which a sediment will occupy whose particles themselves occupy unit volume. It is to be understood that in packing into the sediment the particles in no wise change their agglomeration or shape, the sediment constituting the smallest volume which such agglomerates can occupy without changing.

The relative sediment volume (S') is expressed as volume per unit volume of solids, while in unit volume of suspension the volume of solid particles is V. The sediment volume in unit volume of suspension is therefore $S'V$, and the volume of free liquid is $(1 - S'V)$.

The equation applicable to any concentration of suspension then becomes $\eta_{sp} = kV/(1 - S'V)$.

To prove the equation by measurements of actual suspensions, it must be cast in a form to permit independent evaluation of the two parameters, k and S'. Such a form is $V/\eta_{sp} = 1/k - S'V/k$. When V/η_{sp} is plotted against V, the graph must be linear, if k and S' are constants, with intercepts equal to $1/k$ and $1/S'$, respectively. All of the data reported are handled in this manner.

For most suspensions, it is common experience that the agglomeration which determines S' changes with the rate of shear. If S' were not a constant for a given suspension, experimental proof of the relationship would be excessively difficult. For this reason glass spheres were chosen, since the assumption can reasonably be made that they will remain completely disaggregated. The value of S' also depends upon the particle-size distribution, but it will remain constant for any particular collection of spheres.

EXPERIMENTAL METHOD

The viscosity measurements were made on a rotational viscosimeter designed and manufactured by the Sun Chemical Company, New York, and described by Buchdahl *et al.* (3). The viscosimeter was modified by the substitution of helices of music wire in place of straight torsion wires, with the attendant advantages of great range of deflection without danger of inducing a permanent set in the torsion device. The ends of the helix were tied together with a thread, coaxial with the helix, to hold the weight of the lower suspension off the bearings. The cup has a radius of 0.5994 cm. and a depth of about 8 cm., while the bob has a radius of 0.5499 cm. and an effective length of 4.366 cm. Because of the small clearance between the side walls of cup and bob, the large ratio of length to radius, and the large clearance of the end of the bob from the bottom of the cup, it was concluded that the correction for end effect was negligible (*cf.* Lindsley and Fischer (9)). The speed of rotation is varied through a Graham transmission, whose dial settings are calibrated in R.P.M. The reproducibility of the speed for a given dial setting was within 1 per cent, except at the lowest speed for which data are given, where the reproducibility is less satisfactory. All viscosity values are averages of those computed from data taken at seven rates of shear, equally spaced over the entire range of the instrument. As shown in the tables, the maximum rate of shear is 639 cm./sec./cm. The rate of shear is computed at the center of the annular clearance by multiplying the revolutions per minute of the cup by the instrument constant 1.212. The viscosity is computed by dividing the shearing stress, in dynes per square centimeter, by the shearing rate in centimeters per second per centimeter, the quotient then being in poises.

The viscosimeter is jacketed with a large thermostated bath, filled with transformer oil and maintained at 35.0°C. for all of these experiments. With each suspension, after the material had been placed in the cup and the torsion spring adjusted, measurements were taken at seven points from the lowest to the highest speed of which the viscosimeter is capable, and were again taken at the same points with descending speed. At least three "round trips" from lowest to highest speed and back were made with each suspension, each round trip requiring about 2 min. At least one round trip was required before temperature equilibrium was reached, as shown by declining viscosity during the initial runs. With suspensions of high viscosity, heating of the contents of the cup was noticeable, sufficient to elevate the temperature of the oil bath in the immediate vicinity of the cup a few tenths of a degree. Careful operation was required to estimate, by the fluctuations in viscosity, which values corresponded to 35.0°C. By making the runs at high speeds as brief as possible, and by allowing a period of a few minutes between runs, with the viscosimeter turned off or running slowly, fairly stable values could be obtained.

The glass spheres were prepared by entraining finely powdered and screened Pyrex glass in an oxygen stream supporting the combustion of a gas flame, the flame being contained in a small-diameter pipe connected to a collecting chamber and a water pump in such a way that complete fusion resulted, but the collected powder of spheres was cooled rapidly to prevent sintering. The spheres were cleaned by boiling in a mixture of concentrated nitric and sulfuric acids, and washed by successively stirring with distilled water, settling, and decanting until the supernatant water had the same pH as the distilled water being used. The spheres were separated into two fractions by a sedimentation procedure. The fraction above 10 microns diameter was used in these experiments, in which there were very few individuals above 30 microns, with the most between 10 and 20 microns.

The suspensions were prepared by weighing the desired quantity of glass spheres in a 150-ml. beaker, weighing in the desired quantity of liquid, and stirring for a few minutes with a stirring rod. Weighing was done on a direct reading balance accurate to the nearest 0.1 g., and 20 to 40 g. of each suspension was prepared (7 cc. being required to fill the viscosimeter cup). The greatest concern was for absorption of moisture from the air by those liquids of a hygroscopic nature, so manipulations were conducted with the greatest reasonable dispatch. From the first mixing of the suspension components to the completion of the viscosimeter runs required about 15 min. In all of the liquids used dispersion of the spheres was easy, so nothing more elaborate than the few minutes' stirring with a glass rod was required. Experiments subsequent to those described proved that with smaller spheres de-aeration by evacuation is essential; the erratic results in these data may be due to air bubbles.

After use, the spheres were recovered by repeated washings and decantations with benzene or water, for the types of suspension media used. They were dried on a warm shelf, as it was learned by experience that heating caused the spheres to stick together. After many uses the spheres darkened, presumably owing to the abrasion of steel from the viscosimeter. The dark color was removed by treatment with a mixture of concentrated nitric and sulfuric acids.

In order to calculate the volume concentration from the weight concentration, accurate values of the densities of the suspension media were necessary. These were obtained by measurement of the density using a pycnometer method at 35.0°C. Assuming that there is no swelling of the spheres, which is certainly

TABLE 1

Glass spheres in S.A.E. No. 30 motor oil

Density of oil = 0.8830 g./cc. at 35.0°C.

$100c$	$100V$	VISCOSITY η	$\eta - \eta_0$	$\frac{\eta_0 V}{\eta - \eta_0}$
per cent by weight	*per cent by volume*	*poises*		
0.0	0.0	1.25		
20.0	9.1	1.62	0.37	0.306
33.3	16.5	2.25	1.00	0.206
40.0	20.8	2.46	1.21	0.215
50.0	28.4	3.85	2.60	0.136
60.0	37.3	6.46	5.21	0.090
65.0	42.4	8.40	7.15	0.074
72.0	50.4	23.6	22.3	0.028

TABLE 2

Glass spheres in S.A.E. No. 30 motor oil

RATE OF SHEAR D	VISCOSITY IN POISES							
	Concentration by weight in S.A.E. 30 motor oil (per cent)							
	0.0	20.0	33.3	40.0	50.0	60.0	65.0	72.0
101	1.22	1.595	2.97	2.43	3.80	6.92	9.58	25.4
190	1.25	1.658	2.26	2.47	3.92	6.51	8.30	24.5
274	1.26	1.625	2.27	2.44	3.90	6.28	8.24	23.9
363	1.25	1.608	2.24	2.47	3.83	6.18	8.16	23.5
453	1.25	1.628	2.25	2.48	3.73	6.27	8.40	22.9
546	1.26	1.603	2.24	2.46	3.68	6.49	8.40	22.6
639	1.26	1.585	2.21	2.48	3.72	6.56	8.50	22.2
Average....	1.25	1.617	2.35	2.46	3.85	6.46	8.40	23.6

the case in the systems studied, the volume concentration is calculated from the weight concentration by the formula: $1/V = ((1 - c)/c)(d_s/d_l) + 1$. The density of the Pyrex glass (d_s) is 2.23, and the density of the suspension medium is d_l.

RESULTS

The complete results of measurements of the viscosity of suspensions of glass spheres in two motor oils, castor oil, polyethylene glycol, corn syrup, and sucrose solution are presented in tables 1 to 12. The liquids were selected because of

TABLE 3

Glass spheres in S.A.E. No. 50 motor oil

Density of oil = 0.8897 g./cc. at 35.0°C.

$100c$	$100V$	VISCOSITY η	$\eta - \eta_0$	$\frac{\eta_0 V}{\eta - \eta_0}$
per cent by weight	*per cent by volume*	*poises*		
0.0	0.0	2.31		
20.0	9.1	2.94	0.63	0.332
33.3	16.6	4.11	1.80	0.213
40.0	21.0	4.76	2.45	0.198
50.0	28.5	6.55	4.24	0.155
60.0	37.4	12.71	10.40	0.083
65.0	42.5	17.28	14.97	0.065
68.0	45.9	23.4	21.1	0.045

TABLE 4

Glass spheres in S.A.E. No. 50 motor oil

RATE OF SHEAR D	VISCOSITY IN POISES							
	Concentration by weight in S.A.E. 50 motor oil (per cent)							
	0.0	20.0	33.3	40.0	50.0	60.0	65.0	68.0
101	2.28	3.04	4.18	4.79	6.61	13.08	17.78	25.2
190	2.30	2.97	4.12	4.81	6.46	12.53	17.10	23.8
274	2.30	2.97	4.09	4.79	6.56	12.51	17.16	23.15
363	2.26	2.94	4.08	4.76	6.45	12.55	17.10	22.9
453	2.27	2.92	4.12	4.73	6.52	12.69	17.06	22.9
546	2.26	2.87	4.09	4.66	6.60	12.80	17.18	22.8
639	2.26	2.84	4.11	4.79	6.66	12.86	17.58	23.15
Average . . .	2.28	2.94	4.13	4.76	6.55	12.71	17.28	23.4

TABLE 5

Glass spheres in castor oil

Density of oil = 0.9521 g./cc. at 35.0°C.

$100c$	$100V$	VISCOSITY η	$\eta - \eta_0$	$\frac{\eta_0 V}{\eta - \eta_0}$
per cent by weight	*per cent by volume*	*poises*		
0.0	0.0	3.55		
33.3	17.6	6.45	2.90	0.216
40.0	22.2	7.57	4.02	0.196
50.0	30.0	11.06	7.51	0.142
60.0	40.9	20.15	16.60	0.087
65.0	44.3	34.2	30.7	0.051

their demonstrated Newtonian flow characteristics and because their viscosities were high enough to make the spheres settle at a rate negligible for the measurements being made. Furthermore, it was desired to have liquids representing a

TABLE 6

Glass spheres in castor oil

RATE OF SHEAR D	VISCOSITY IN POISES					
	Concentration by weight in castor oil (per cent)					
	0.0	33.3	40.0	50.0	60.0	65.0
101	3.57	6.61	7.76	11.15	21.1	37.2
190	3.60	6.51	7.48	10.92	20.35	35.9
274	3.59	6.50	9.45	11.03	20.35	35.6
363	3.55	6.41	7.51	11.08	20.1	33.0
453	3.62	6.40	7.58	11.03	19.95	33.4
546	3.51	6.39	7.59	11.02	19.75	32.1
639	3.50	6.36	7.64	11.18	19.45	32.1
Average....	3.56 (3.51)* (3.59)*	6.45	7.57	11.06	20.15	34.2

* Values in parentheses are averages of repeated determinations using fresh samples.

TABLE 7

Glass spheres in glucose corn syrup

Density of solution = 1.3338 g./cc. at 35.0°C.

$100c$	$100V$	VISCOSITY η	$\eta - \eta_0$	$\frac{\eta_0 V}{\eta - \eta_0}$
per cent by weight	*per cent by volume*	*poises*		
0.0	0.0	2.30		
20.0	13.05	3.89	1.59	0.189
33.3	23.2	6.01	3.71	0.144
50.0	37.4	13.27	10.97	0.078
60.0	47.4	32.8	30.5	0.036

TABLE 8

Glass spheres in glucose corn syrup

RATE OF SHEAR D	VISCOSITY IN POISES				
	Concentration by weight in corn syrup (per cent)				
	0.0	20.0	33.3	50.0	60.0
101	2.43	4.10	6.31	13.58	35.1
190	2.34	3.92	6.22	13.23	34.9
274	2.35	3.92	6.05	13.34	34.5
363	2.33	3.87	5.88	13.36	33.7
453	2.31	3.84	5.86	13.27	30.1
546	2.31	3.81	5.84	13.16	31.4
639	2.29	3.76	5.80	13.04	30.2
Average.......	2.34 (2.26)*	3.89	5.98	13.27	32.8

* Values in parentheses are averages of repeated determinations using fresh samples.

wide variety of chemical types. The aqueous solutions presented the greatest difficulty, since solutions of high viscosity necessitated high concentrations,

TABLE 9

Glass spheres in polyethylene glycol

Density of liquid = 1.1529 g./cc. at 35.0°C.

$100c$	$100V$	VISCOSITY η	$\eta - \eta_0$	$\frac{\eta_0 V}{\eta - \eta_0}$
per cent by weight	*per cent by volume*	*poises*		
0.0	0.0	0.268		
20.0	0.1143	0.454	0.186	0.165
33.3	0.206	0.698	0.430	0.128
50.0	0.342	1.377	1.109	0.083
65.0	0.490	5.07	4.80	0.027

TABLE 10

Glass spheres in polyethylene glycol

RATE OF SHEAR D	VISCOSITY IN POISES				
	Concentration by weight in polyethylene glycol (per cent)				
	0.0	20.0	33.3	50.0	65.0
101	0.1520	0.456	0.760	1.443	5.01
190	0.242	0.445	0.686	1.373	5.16
274	0.252	0.449	0.673	1.345	5.13
363	0.275	0.465	0.698	1.375	5.10
453	0.272	0.441	0.694	1.373	5.05
546	0.282	0.464	0.689	1.363	5.04
639	0.288	0.456	0.685	1.370	5.00
Average.......	0.268	0.454	0.698	1.377	5.07

TABLE 11

Glass spheres in sucrose solution

Density of solution = 1.3178 g./cc. at 35.0°C.

$100c$	$100V$	VISCOSITY η	$\eta - \eta_0$	$\frac{\eta_0 V}{\eta - \eta_0}$
per cent by weight	*per cent by volume*	*poises*		
0.0	0.0	0.792		
20.0	12.9	1.345	0.553	0.185
33.3	22.8	2.34	1.55	0.116
40.0	28.2	2.95	2.16	0.103
60.0	47.0	11.59	10.80	0.0345

which in turn created the problem of hygroscopic behavior or, alternately, meant solutions of materials having high molecular weight which will degrade under high shearing stress. The selection of sucrose and corn syrup represents workable

TABLE 12

Glass spheres in sucrose solution

RATE OF SHEAR D	VISCOSITY IN POISES				
	Concentration by weight in sucrose solution (per cent)				
	0.0	20.0	33.3	40.0	60.0
101	0.835	1.370	2.74	3.50	11.69
190	0.808	1.375	2.46	3.15	11.47
274	0.784	1.345	2.35	2.97	11.48
363	0.782	1.353	2.26	2.88	11.69
453	0.780	1.340	2.24	2.78	11.83
546	0.773	1.322	2.18	2.73	11.63
639	0.781	1.310	2.12	2.66	11.36
Average.......	0.792	1.345	2.34	2.95	11.59

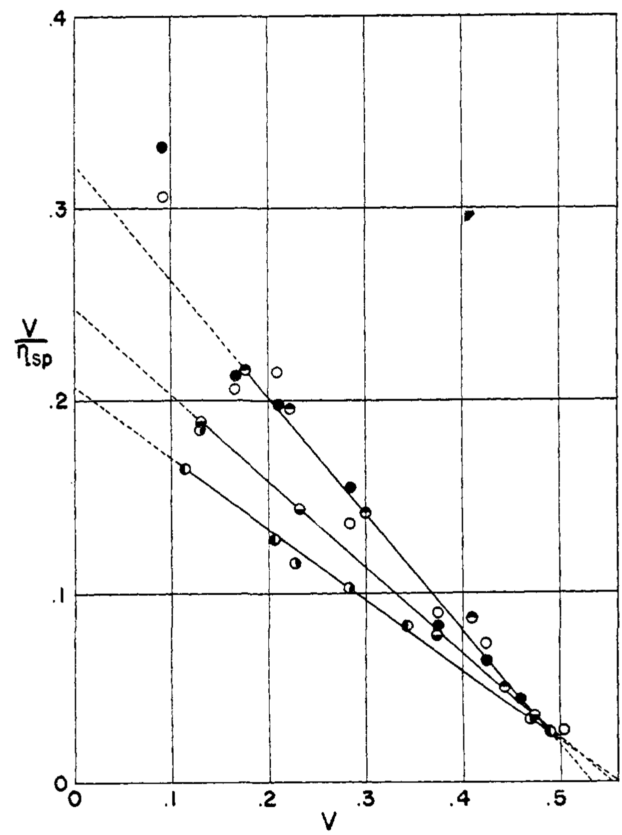

FIG. 1. Plot of V/η_{sp} *vs.* V for suspensions of glass spheres in various media: ○, S.A.E. No. 30 motor oil; ●, S.A.E. No. 50 motor oil; ◓, castor oil; ◒, corn syrup; ◐, polyethylene glycol; ◑, sucrose solution.

compromises in both respects, but requires that exposure to the atmosphere be kept to a minimum. The corn syrup used is the product made by the acid conversion of corn starch and sold commercially as glucose.

The data contained in the tables are shown graphically in figure 1, in which V/η_{sp} is plotted against V. Several conclusions are immediately apparent:

(*1*) Except for a few points believed to be in error, the data fall on straight lines, indicating that the form of the relationship is valid over the range of concentration tested.

(*2*) For all of the six liquids tested, the intercepts of the linear graphs on the V-axis ($1/S'$) vary only slightly, indicating that S' is a constant depending primarily upon the suspended solid.

(*3*) There are three linear graphs resulting from the measurements on the six liquids, with approximately the same V-axis intercept but with three V/η_{sp}-axis intercepts. The latter intercept is $1/k$, indicating that the value of k depends upon the suspension medium. Furthermore, the data from chemically similar liquids fall upon the same line, indicating that k is related to properties of the liquids other than their viscosities. Two motor oils of widely different viscosity and castor oil have one value of k, the two aqueous solutions have another value, and polyethylene glycol has the third value of k.

The variation of viscosity with the rate of shear is indicated in the tables. In most of the cases reported there is no systematic variation with rate of shear, the fluctuations being an indication of the experimental reproducibility. There is some indication with the high-molecular-weight substances, polyethylene glycol and corn syrup, and with the hygroscopic substances, sucrose solution and corn syrup, that the viscosity tends to drop slowly and continuously. However, these deviations are small and are not regarded as evidence for non-Newtonian characteristics of flow, e.g., thixotropy, but rather as due either to the slow absorption of moisture or to the slow decrease in molecular weight due to mechanical stress. The assumption of a single viscosity for each suspension is made on the basis of these data.

DISCUSSION

As shown on the accompanying graph (figure 1) the two points for suspensions of spheres in motor oils at a concentration below 10 per cent by volume do not come near the straight line. At low suspension concentrations the suspension has nearly the same viscosity as the medium, and since the specific viscosity contains the difference of the two viscosities, a small error in either results in a large error in the specific viscosity. However, this limitation is not believed to affect the validity of the proposed equation at low suspension concentrations. As the suspension concentration becomes smaller, the relationship $\eta_{sp} = kV/(1 - S'V)$ approaches the Einstein approximation within a few per cent, since $(1 - S'V)$ approaches 1. The experiments of Eirich, Bunzl, and Margaretha (5) indicate that suspensions of spheres behave concordantly with the Einstein equation at low concentration, in the range in which the two relationships are equivalent in form. On the basis of the experimental data presented, and the reduction to the Einstein equation at low concentrations, the conclusion is reached that the proposed equation is valid for suspensions of inert spheres at any concentration.

To test the proposed equation with independently obtained data, the work of Vand (11) was utilized. His data were used in the form as taken, without the corrections he applied, and the results are shown in figure 2, the original data and the calculations from them being shown in tables 13 to 15. The Ostwald visco-

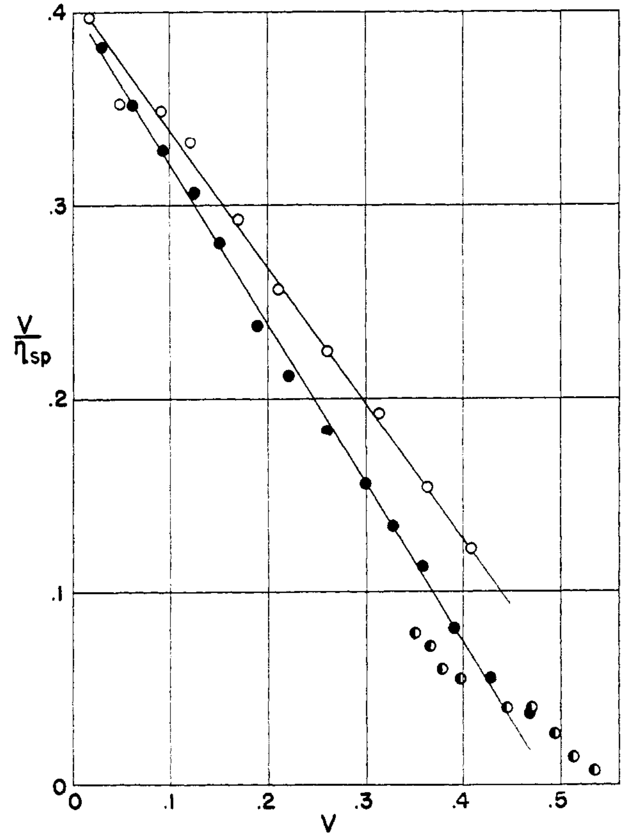

FIG. 2. Plot of V/η_{sp} vs. V for suspensions of glass spheres in zinc iodide–glycerol solution (from the data of Vand (11)).

simeters both give linear graphs when V/η_{sp} is plotted against V. However, there is a difference between the two viscosimeters. The data from the rotational viscosimeter show no correlation, presumably because the instrument was constructed with large clearance between cup and bob and with large diameter compared to the depth, causing the errors discussed by Fischer and Lindsley (6). Analysis of the data obtained with the capillary viscosimeters is beyond the scope of this paper, other than to demonstrate that the data fit the form of equation proposed. In connection with the analysis of the Vand data, recalculation was made to determine the fit with the equations he proposes. These results are

TABLE 13

Vand data obtained with Ostwald viscosimeter No. 3 on suspensions of glass spheres

(1)* CORRECTED VOLUME CONCENTRATION c	(2)* $1 - \eta_0/\eta$ (UNCORRECTED)	(3) RELATIVE VISCOSITY (UNCORRECTED)	(4)* $1 - \eta_0/\eta$ (CORRECTED)	(5) RELATIVE VISCOSITY (CORRECTED)	(6) CALCULATED RELATIVE VISCOSITY Derived	(7) CALCULATED RELATIVE VISCOSITY Empirical	(8)* VOLUME CONCENTRATION (UNCORRECTED) V	(9) $\frac{\eta_0 V}{\eta - \eta_0}$
0.01942	0.042	1.042	0.0541	1.058	1.051	1.051	0.01742	0.397
0.05375	0.120	1.135	0.1545	1.183	1.156	1.158	0.0482	0.353
0.10035	0.205	1.255	0.2640	1.359	1.338	1.354	0.0900	0.349
0.1332	0.270	1.370	0.3478	1.532	1.463	1.499	0.1194	0.333
0.1887	0.366	1.577	0.4720	1.895	1.733	1.836	0.1692	0.293
0.2342	0.450	1.816	0.5790	2.38	1.989	2.00	0.2100	0.257
0.2892	0.535	2.145	0.6890	3.22	2.337	2.36	0.2595	0.225
0.3516	0.622	2.64	0.8010	5.03	2.787	2.83	0.3155	0.192
0.4045	0.703	3.365	0.9050	10.53	3.210	3.29	0.3630	0.1535
0.4550	0.770	4.35	0.9920	12.50	3.660	3.78	0.4080	0.1220

* From Vand (reference 11, Part II, table 1, page 305).

Column 3 calculated from column 2, and column 5 from column 4.

Column 6 calculated from $\eta_r = 1 + 2.5c + 7.349c^2$.

Column 7 calculated from $\eta_r = 1 + 2.5c + 7.17c^2 + 16.2c^3$.

Column 9 calculated from column 8 divided by the product of column 2 times column 3.

TABLE 14

Vand data obtained with Ostwald viscosimeter No. 4 on suspensions of glass spheres

(1)* CORRECTED VOLUME CONCENTRATION c	(2)* $1 - \eta_0/\eta$ (UNCORRECTED)	(3) RELATIVE VISCOSITY (UNCORRECTED)	(4)* $1 - \eta_0/\eta$ (CORRECTED)	(5) RELATIVE VISCOSITY (CORRECTED)	(6) CALCULATED RELATIVE VISCOSITY Derived	(7) CALCULATED RELATIVE VISCOSITY Empirical	(8)* VOLUME CONCENTRATION (UNCORRECTED) V	(9) $\frac{\eta_0 V}{\eta - \eta_0}$
0.0326	0.074	1.080	0.0861	1.095	1.089	1.090	0.03055	0.382
0.0652	0.148	1.175	0.1722	1.209	1.194	1.198	0.0611	0.352
0.0978	0.218	1.283	0.254	1.340	1.314	1.328	0.0916	0.329
0.1328	0.288	1.403	0.335	1.503	1.471	1.500	0.1242	0.307
0.1609	0.349	1.535	0.406	1.683	1.591	1.654	0.1506	0.281
0.2013	0.443	1.798	0.516	2.07	1.801	1.925	0.1886	0.238
0.2360	0.511	2.04	0.594	2.46	2.00	2.20	0.2210	0.212
0.2781	0.587	2.42	0.683	3.06	2.26	2.60	0.2605	0.1830
0.3197	0.657	2.92	0.764	4.24	2.55	3.06	0.2995	0.1560
0.3502	0.712	3.47	0.828	5.81	2.77	3.44	0.3280	0.1328
0.3822	0.7598	4.16	0.884	8.62	3.03	3.91	0.3580	0.1132
0.4170	0.8280	5.81	0.963	27.0	3.32	4.47	0.3908	0.0812
0.4565	0.8885	8.98	1.032	Negative	3.68	5.19	0.4275	0.0556
0.5000	0.9266	13.65	1.078	Negative	4.09	6.08	0.4680	0.0370

* From Vand (reference 11, Part II, table 2, page 305).

Column 3 calculated from column 2, and column 5 from column 4.

Column 6 calculated from $\eta_r = 1 + 2.5c + 7.349c^2$.

Column 7 calculated from $\eta_r = 1 + 2.5c + 7.17c^2 + 16.2c^3$.

Column 9 calculated from column 8 divided by the product of column 2 times column 3.

contained in three self-explanatory tables (tables 13 to 15); the comparison of columns 3 or 5 with columns 6 or 7 constitutes the basis for the assertion that the fit is satisfactory only up to a concentration of 25 per cent by volume.

A check of the physical significance of the parameter S' was made by comparing the values obtained from the graph of the viscosity data with the value obtained by direct measurement of the sediment volume. For the latter, 10 g. of the powder of glass spheres was weighed into a 10-cc. graduated cylinder, and water was added to permit the particles to settle freely. Centrifuging at an acceleration about 700 times that of gravity did not settle the spheres further. The observed volume was 8.8 cc., giving a specific volume of 0.88 cc. per gram. The density of Pyrex glass is 2.23, from which the relative sediment volume is computed to be

TABLE 15

Vand data obtained with Couette viscosimeter on suspensions of glass spheres

(1)* Corrected volume concentration c	(2)* $1 - \eta_0/\eta$ (uncorrected)	(3) Relative viscosity (uncorrected)	(4)* $1 - \eta_0/\eta$ (corrected)	(5) Relative viscosity (corrected)	(6) Calculated relative viscosity: Derived	(7) Calculated relative viscosity: Empirical	(8)* Volume concentration (uncorrected) Γ	(9) $\frac{\eta_0\Gamma}{\eta - \eta_0}$
0.355	0.8059	5.14	0.827	5.79	2.81	3.52	0.355	0.0790
0.371	0.835	6.05	0.857	7.00	2.94	3.75	0.366	0.0723
0.383	0.862	7.23	0.886	8.77	3.00	3.92	0.378	0.0605
0.402	0.879	8.27	0.904	10.42	3.20	4.23	0.397	0.0546
0.451	0.918	12.18	0.944	17.87	3.62	5.08	0.445	0.0398
0.476	0.9205	12.58	0.946	18.50	3.85	5.57	0.470	0.0406
0.500	0.948	19.27	0.974	38.5	4.09	6.07	0.493	0.0270
0.520	0.9710	34.5	0.997	334.	4.29	6.53	0.513	0.0153
0.541	0.9728	36.8	0.998	500.	4.49	7.00	0.534	0.0149

* From Vand (reference 11, Part II, table 3, page 309).
Column 3 calculated from column 2, and column 5 from column 4.
Column 6 calculated from $\eta_r = 1 + 2.5c + 7.349c^2$.
Column 7 calculated from $\eta_r = 1 + 2.5c + 7.17c^2 + 16.2c^3$.
Column 9 calculated from column 8 divided by the product of column 2 times column 3.

1.96 cc. per cubic centimeter. The values from the graphical intercepts are 1.77, 1.81, and 1.88, respectively.

The above experimental check indicates that the effective volume of the suspended particles, which determines the suspension viscosity, is closely related to the relative sediment volume; by postulate, the two are identical. By implication in the derivation, S' varies only with the particle aggregation, and for the systems used such seems to be the case. However, this restriction is not necessary, and S' will be increased if the particles collect shells of material around them in their motion through a suspension medium containing large molecules. The absence of such an effect with the spheres used is probably due to their large size relative to molecular dimensions.

The extension of the equation to suspensions of practical significance is complicated by the fact that S' generally will not be a constant for a given suspension, but will depend on the past history of the suspension and upon the shear to which it is being subjected. The hypothesis is advanced that the relationship is valid for any suspension, and affords a means for calculating the relative sedimentation volume of the suspended solid for the specific conditions under which viscosity measurements are made.

The extension of the equation to suspensions of macromolecules should be similar and complicated by the same type of phenomenon. The parameter S' should then be effective molecular size, that is, relative molecular packing in the same sense as relative sediment volume, with the complication that the molecular association responsible for the thixotropy of solutions of macromolecules changes analogously to the aggregation of the particles of a suspension, with a corresponding change in S'.

The physical significance of k is conjectural. The independence of k and S' shown by the particular systems chosen is unexpected and not to be anticipated generally. The parameter k may be analogous to a frictional coefficient, and therefore is influenced by factors such as particle roughness and shape and the presence or absence of shells around the particles, derived from the suspension medium. These factors in turn are influenced by the particle size and particle-size distribution which determine S'. The fact that a single dispersed material, as herein, demonstrates a nearly constant value of S' and a changing value of k as the suspension medium changes indicates that the "frictional coefficient" can be changed without the formation of a film around the spheres of sufficient thickness to greatly affect S'.

The intercept of the graph of V/η_{sp} against V on the V-axis represents the point at which $S'V$ equals unity. It is easily possible to make up a suspension in which the proportions are such that V is greater than the intercept value. There is then insufficient liquid to fill the voids between particles. The work of shearing may be sufficient to change S' and therefore move the V-axis intercept, or some of the voids will be filled with air and another phase introduced into the system. The appearance of scattered points at values of V greater than the V-axis intercept is therefore to be expected.

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THE INVESTIGATION OF THE PROPERTIES OF NITROCELLULOSE MOLECULES IN SOLUTION BY LIGHT-SCATTERING METHODS. II

EXPERIMENTAL RESULTS AND INTERPRETATION[1]

RICHARD M. BADGER AND ROBERT H. BLAKER

Gates and Crellin Laboratories of Chemistry, California Institute of Technology, Pasadena, California

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INTRODUCTION

Techniques and instruments developed in this laboratory for the determination of the sizes and shapes of high-polymer molecules by the light-scattering method were described in a previous paper (2), which in the following will be referred to as Paper No. I. The present paper describes the application of these methods to a study of a series of nitrocellulose fractions which has resulted in rather definite conclusions regarding the character of the nitrocellulose molecule in solution. The results obtained from the light-scattering measurements are correlated with viscosity and diffusion data by the use of recently developed theories.

EXPERIMENTAL

Materials

The nitrocellulose specimens employed in this investigation include several unfractionated commercial materials and seven fractions. Fractions S-4,3, S-3,4, S-1,1-4, P-3,2, and P-4,2, all of approximately 13.4 per cent nitrogen content, were provided by the courtesy of Professor J. W. Williams of the University of Wisconsin.[2] Fractions designated as A and B were kindly supplied by Dr. R. L. Mitchell of Rayonnier, Inc. The solvent used in all experiments was chemically pure acetone.

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[2]The details of the fractionation procedure are described in O.S.R.D Report No. 4123, PBA No. 18861, entitled "The Characterization and Solubility of Fractionated Wood Pulp and Cotton Linters Nitrocelluloses," by J. W. Williams *et al.*